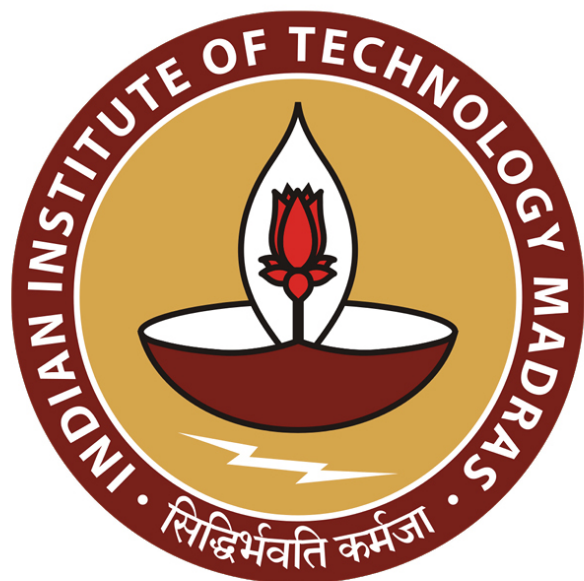


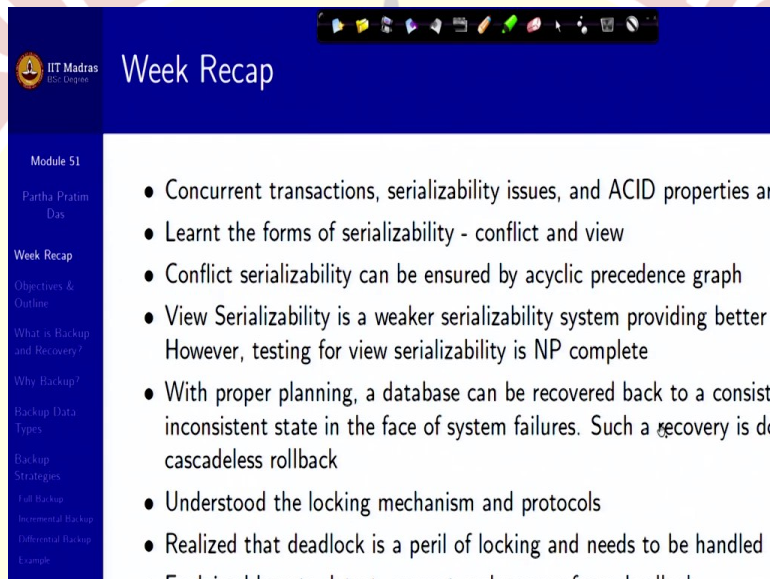


IIT Madras
ONLINE DEGREE

Database Management System
Professor. Partha Pratim Das
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur
Lecture No. 01
Backup & Recovery/1: Backup/1

Welcome to module 51, Week 11 of Database Management Systems course in the IIT, Madras online B.Sc. program.

(Refer Slide Time: 0:30)



Week Recap

- Concurrent transactions, serializability issues, and ACID properties are
- Learnt the forms of serializability - conflict and view
- Conflict serializability can be ensured by acyclic precedence graph
- View Serializability is a weaker serializability system providing better performance. However, testing for view serializability is NP complete
- With proper planning, a database can be recovered back to a consistent state in the face of system failures. Such a recovery is do cascadeless rollback
- Understood the locking mechanism and protocols
- Realized that deadlock is a peril of locking and needs to be handled

In the last week we have talked extensively about transactions, that is how to perform tasks, various types of tasks in the databases, and what are the desired properties of atomicity, consistency, isolation and durability that we need to ensure in a transaction to keep the database consistent and everything going perfect.

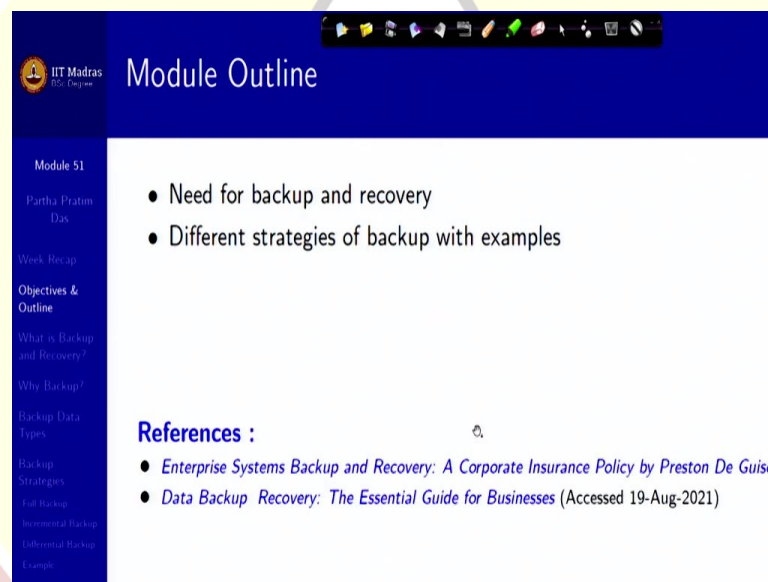
When transactions get concurrent, which they frequently do, we get into serializability issues because instructions from two or more transactions are getting mixed up in the execution order, and we need to ensure correctness in spite of that and there are different serializability concepts, conflict serializability we have shown can be ensured by acyclic precedence graph which is computable in polynomial time.

Whereas a weaker form of serializability, which gives us more range of serial schedules, needs a NP complete algorithm, or rather, rather it has NP completeness. So it cannot, is not likely to

have a polynomial time algorithm. So we use certain adhoc schemes and with proper planning we can recover a database to a consistent state from an inconsistent events in case of failures. We have looked at cascaded, as well as cascadeless rollback for such kind of failures in transactions.

Finally, we have looked at a locking mechanism to ensure that instead of pre-deciding on serializable schedules, it is possible to have schedules, concurrent schedules with better throughput be generated if by maintaining certain minimum criterion for isolation. And we have seen that deadlock happens to be a big disadvantage in that whole process. So, we have explained how to detect, prevent, and recover from deadlock including time based protocols that avoid deadlocks.

(Refer Slide Time: 3:02)



The screenshot shows a presentation slide titled "Module Outline" from IIT Madras. The slide is divided into a left sidebar and a main content area. The sidebar lists the following items: "Module 51", "Partha Pratim Das", "Week Recap", "Objectives & Outline", "What is Backup and Recovery?", "Why Backup?", "Backup Data Types", "Backup Strategies", "Full Backup", "Incremental Backup", "Differential Backup", and "Example". The main content area contains the following text:

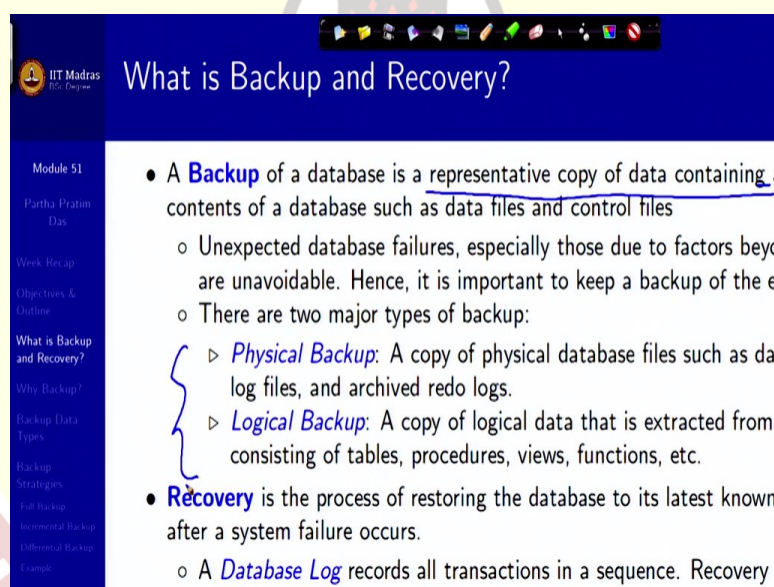
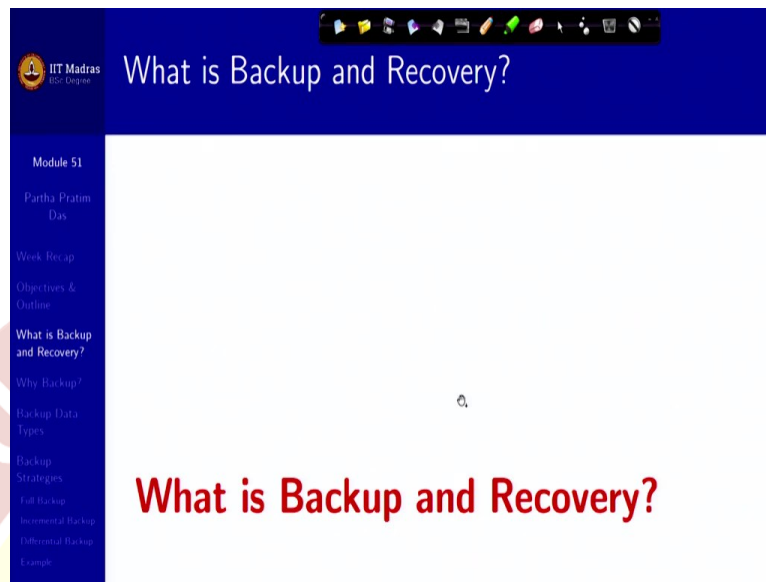
- Need for backup and recovery
- Different strategies of backup with examples

References :

- *Enterprise Systems Backup and Recovery: A Corporate Insurance Policy* by Preston De Guise
- *Data Backup Recovery: The Essential Guide for Businesses* (Accessed 19-Aug-2021)

So, with that with that background ready with us we will now move into discussing other very important factors in terms of database management. These are about backup and recovery. So, we start discussing in this module about the need for backup and recovery, and some of the basic strategies for backup.

(Refer Slide Time: 3:20)





What is Backup and Recovery?

- Module 51
 - Partha Pratim Das
 - Week Backup
 - Objectives & Outline
 - What is Backup and Recovery?
 - Why Backup?
 - Backup Data Types
 - Backup Strategies
 - Full Backup
 - Incremental Backup
 - Differential Backup
 - Example

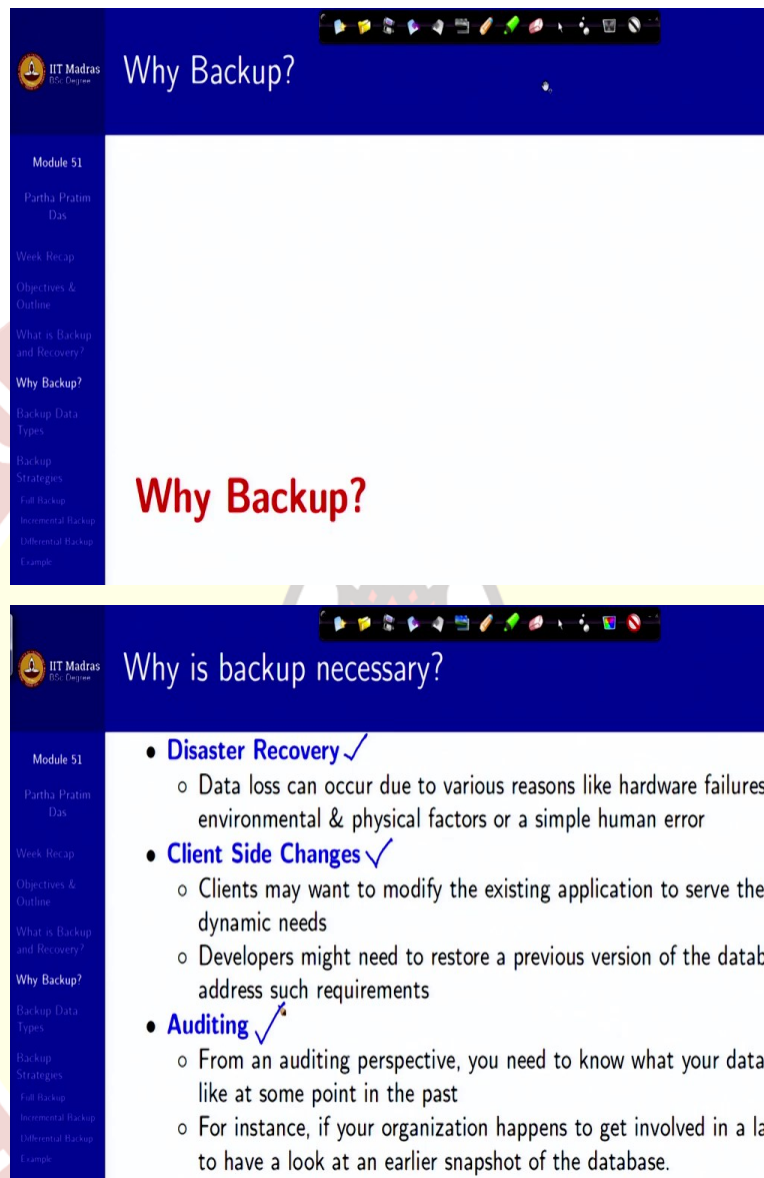
- A **Backup** of a database is a representative copy of data containing : contents of a database such as data files and control files
 - Unexpected database failures, especially those due to factors beyo are unavoidable. Hence, it is important to keep a backup of the e
 - There are two major types of backup:
 - ▷ **Physical Backup**: A copy of physical database files such as da log files, and archived redo logs.
 - ▷ **Logical Backup**: A copy of logical data that is extracted from consisting of tables, procedures, views, functions, etc.
- **Recovery** is the process of restoring the database to its latest known after a system failure occurs.
 - A **Database Log** records all transactions in a sequence. Recovery

Now why do we need backup? Certainly, we all know the importance of backup. It's not, it will not have to be a database for which we have to specifically look for backup, we all backup our files, our file systems and if we fail to do that then we lose them. So, backup for a, of a database is similarly, like as we do is a representative copy of the data containing all the necessary contents of the database like, data tables, control files so that unexpected database failure, which are due to factors that are beyond our control can still be handled by recovering from those backups.

There are primarily two kinds of backup that we deal with. One is a physical backup where you just copy everything physically, like data control files, log files, archives and so on. Other is a logical backup, that instead of physically copying everything you can extract the logical data, may be some kind of an export consisting of tables and procedures and views and so on and back them up. So both strategies, depending on the context, can be used.

Now, given that we are doing a backup, certainly a, the recovery process will also have to be decided. So, what is a recovery? Recovery is the process of restoring the database to its latest known consistent state after a system failure occurs at some random point of time. And a database log, we will talk about this in lot more detail, a database log records all transactions in a sequence. As transactions are done, they are logged, kind of written down. So, recovery using logs is quite popular in databases and we will try to, we will take look into those.

(Refer Slide Time: 5:24)



Why Backup?

Module 51
Partha Pratim Das
Week Recap
Objectives & Outline
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Why is backup necessary?

- **Disaster Recovery** ✓
 - Data loss can occur due to various reasons like hardware failures, environmental & physical factors or a simple human error
- **Client Side Changes** ✓
 - Clients may want to modify the existing application to serve their dynamic needs
 - Developers might need to restore a previous version of the database to address such requirements
- **Auditing** ✓
 - From an auditing perspective, you need to know what your data looked like at some point in the past
 - For instance, if your organization happens to get involved in a lawsuit, you need to have a look at an earlier snapshot of the database.

So this is, what are the types, why backup is necessary? So what are the different situations for which you will do backup. So, depending on your requirement in future the nature of the backup could also be different. We just explained that the need for backup for disaster recovery, which is quite obvious. But there may be needs in according to the basic business process changes. Client want to modify the existing application to serve the business needs that have evolved at a later point of time.

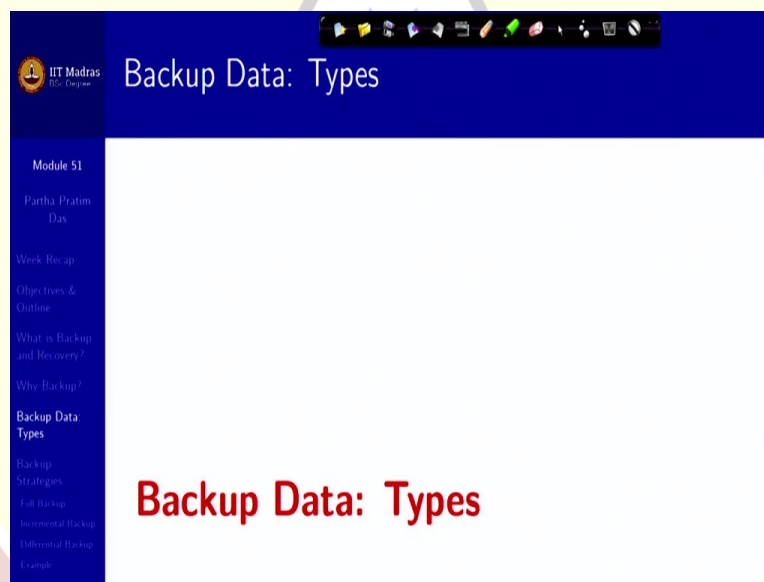
So, the developers might need to restore a previous version of the database before these changes, in order to address such requirements or keep consistency or make sure that you can guarantee

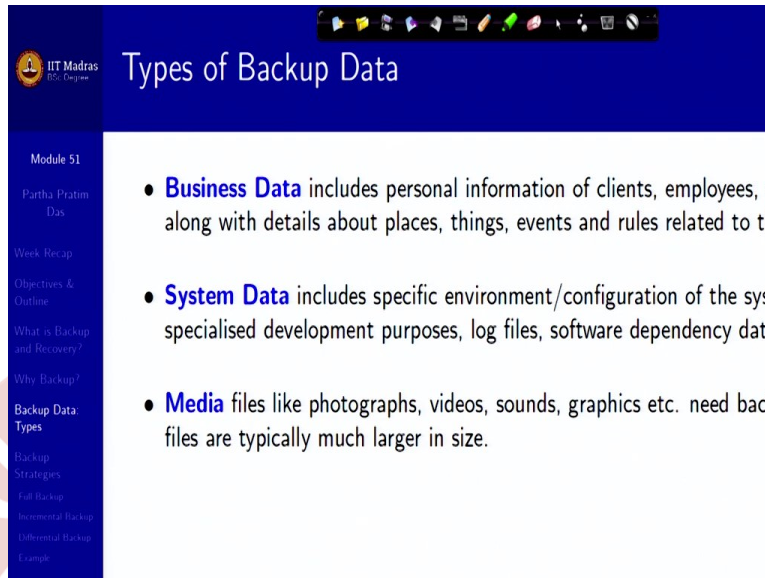
that your changes have not regressed in terms of what you used to do correctly and so on. Auditing becomes another important factor.

You have seen that several financial misappropriation, fraud is going around all across the world and so from audited perspective you need to have a look at the data, and the database organization in the past so that if your organization is involved in a lawsuit you may want to have a look at the earlier snapshot of the database. This needs backups to be there.

And certainly, the downtime, without a backup the downtime of a, of a database system would be much higher. And downtime is certainly not at all good for a business. So you will want to minimize that downtime.

(Refer Slide Time: 7:15)





IIT Madras
Types of Backup Data

Module 51
Partha Pratim Das

- **Business Data** includes personal information of clients, employees, c along with details about places, things, events and rules related to th
- **System Data** includes specific environment/configuration of the syst specialised development purposes, log files, software dependency data
- **Media** files like photographs, videos, sounds, graphics etc. need back files are typically much larger in size.

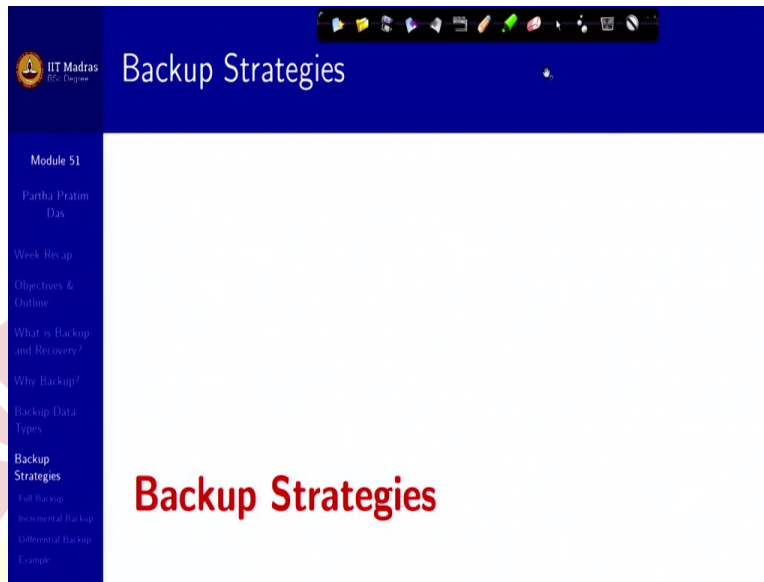
Week Recap
Objectives & Outline
What is Backup and Recovery?
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So when I do, we do backup, let us see what are the types of data that we would backup. Certainly, we will backup the business data. The business data is what primarily we have designed for, the information of clients, employees, contractors, inventories, sales information, course details, etcetera, etcetera, all the rules and all that. That is the business data. That is, that is the basic model that you have created through all these design and so on.

The second is, you will also need to backup the system data. The specific environment, the configuration, the systems catalog that you had for the database, the log files, software dependency, different disk images and so on. So, this is a second category of data.

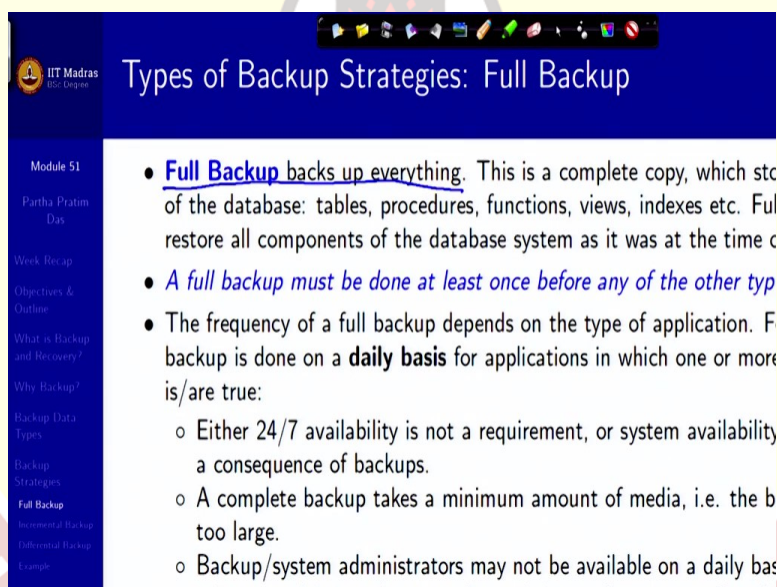
And the third category of data are typically called the media data. The photographs, videos, sound, graphics, all that we have talked about as kind of blob data, that have gone into the database. You had photographs of employees, you had video recordings of important board meetings, and so on so forth, they will also have to be backed up, and typically this media files are much larger in size compared to the typical database files which are mostly textual.

(Refer Slide Time: 8:40)



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Backup Strategies



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Types of Backup Strategies: Full Backup

- **Full Backup** backs up everything. This is a complete copy, which stores of the database: tables, procedures, functions, views, indexes etc. Full restore all components of the database system as it was at the time of backup.
- A full backup must be done at least once before any of the other types.
- The frequency of a full backup depends on the type of application. Full backup is done on a **daily basis** for applications in which one or more of the following is/are true:
 - Either 24/7 availability is not a requirement, or system availability is a consequence of backups.
 - A complete backup takes a minimum amount of media, i.e. the backup is not too large.
 - Backup/system administrators may not be available on a daily basis.

Types of Backup Strategies: Full Backup

- **Full Backup** backs up everything. This is a complete copy, which stores all components of the database system as it was at the time of the backup.
- *A full backup must be done at least once before any of the other types.*
- The frequency of a full backup depends on the type of application. Full backup is done on a daily basis for applications in which one or more of the following is/are true:
 - Either 24/7 availability is not a requirement, or system availability is a consequence of backups.
 - A complete backup takes a minimum amount of media, i.e. the backup is not too large.
 - Backup/system administrators may not be available on a daily basis.

Now, let us look at what could be the strategy to backup. So, the question certainly is, how much do we backup, how often do we backup, what do we backup? So, all these questions are answered in terms of forming a consistent backup strategy for the organization.

So, the first and most important kind of backup is called a Full Backup, which as the name suggests backs up everything. It is a complete copy, it is a complete replica of what you have in the database. Tables, procedures, functions, system views, everything. So a Full Backup can completely restore all (complete) all components of the database system as it was at that time the crash happened.

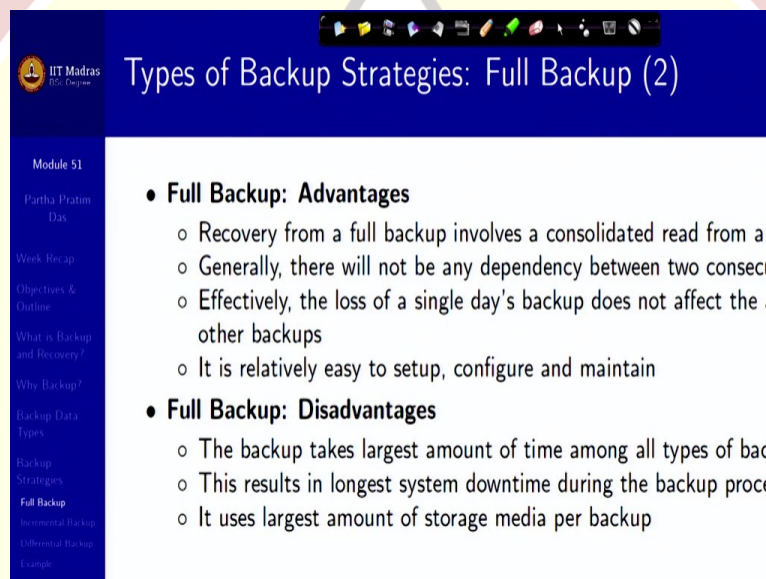
So, what is very important, that when you are going to do something major in your database, or when you are going to do any other kind of backup, take any other kind of backup, before that at least once you must have a full backup. And then you can have anything else. Now naturally the question is, how often would you take full backup? That depends on the type of application. A full backup may be done on a daily basis, where you have a, I mean you do not have a requirement of a 24/7 availability.

See if your system, say, your net banking system, which is 24/7 available. If you are, if users are currently using and changing the table every second, every millisecond, certainly you will not be able to do a full backup easily. Because the system availability is going to be restricted when you do this backup. So, this is one consideration.

Other is, naturally how much media do you need. If you do daily, backup on a daily basis and your backup requirement say, is 2 terabyte, then you can think of that in a year over 365, 366 days how much big media that you will need. Also, you may want to do the backup on a daily basis because you may not have easily available system administrators who can easily restore the system from a not so full backup also.

So, you just want to make sure that full backup is available. So, these are some of the reasons for which you do a full backup. Certainly, the advantage is obvious that it, it is a consolidated read from a single backup.

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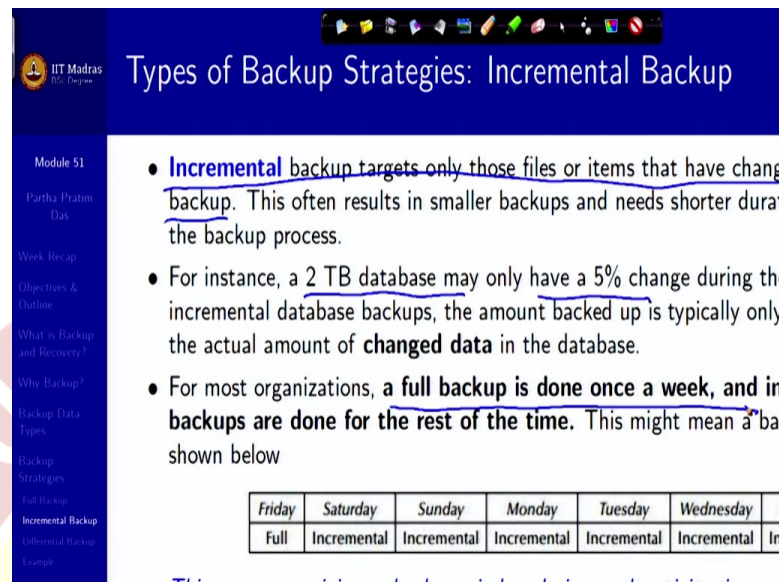


The screenshot shows a presentation slide from IIT Madras. The title is 'Types of Backup Strategies: Full Backup (2)'. The slide is part of Module 51, presented by Partha Pratim Das. The left sidebar lists the following topics: Week Backup, Objectives & Outline, What is Backup and Recovery?, Why Backup?, Backup Data Types, Backup Strategies, Full Backup, Incremental Backup, Differential Backup, and Example. The main content area lists the following points:

- **Full Backup: Advantages**
 - Recovery from a full backup involves a consolidated read from a single backup
 - Generally, there will not be any dependency between two consecutive full backups
 - Effectively, the loss of a single day's backup does not affect the ability to recover other backups
 - It is relatively easy to setup, configure and maintain
- **Full Backup: Disadvantages**
 - The backup takes largest amount of time among all types of backup
 - This results in longest system downtime during the backup process
 - It uses largest amount of storage media per backup

Generally it will not depend on anything else. So, there is two consecutive full backups do not have a dependency between them, and effectively the loss of single days backup does not affect the ability to recover other backups. Easy to set up, configure, maintain, but certainly, it takes a large amount of time, takes a large amount of results in a long down time for the, for the system for the backup process, and needs a huge amount of media. So, we will always not be able to do a full backup.

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Types of Backup Strategies: Incremental Backup

- Incremental backup targets only those files or items that have changed since the last backup. This often results in smaller backups and needs shorter duration for the backup process.
- For instance, a 2 TB database may only have a 5% change during the incremental database backups, the amount backed up is typically only the actual amount of **changed data** in the database.
- For most organizations, a full backup is done once a week, and incremental backups are done for the rest of the time. This might mean a backup schedule shown below

Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday
Full	Incremental	Incremental	Incremental	Incremental	Incremental	Incremental

So incidentally, just to, just to give you an idea, I use a laptop which has a one terabyte hard disk, and I have about 500, about 600 gigabytes of data in that. My, different personal and work data. So, every week I do a backup. Now when I do a backup, it takes about 3 to 4 hours, when I cannot work, when I have to make sure that the power does not go off so that the backup is inconsistent, closed at the middle.

So, what I started using is certain software which compute what has changed from the backup I did last week, and backs up only that part. It just looks at the time stamp of the files. I have to make sure that my system has a correct timestamp always. So, anything that has a change timestamp, which is more than the timestamp in the last backup on my external hard disk is what is backed up. So, that is the basic idea of doing an incremental backup. It backs up only those files and item that has changed since the last backup.

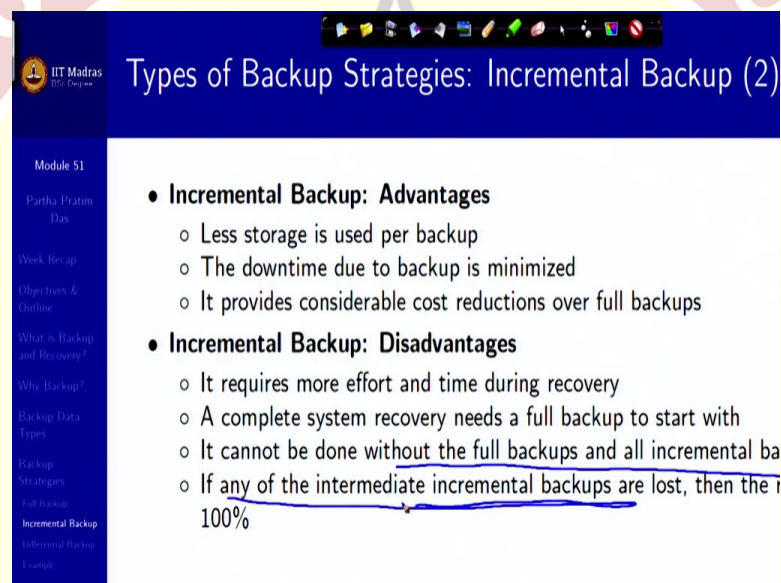
So, when I started doing this, the, the backup requirement fell from 500 600 GB to down to 5, 6 GB. I mean, not even 1 percent, and the time came down to about from 3 to 4 hours, it came down to about 10 to 15 minutes. So, in a typical, maybe 2 TB database also, it may have changed only 5 percent during the day. It, all changes will not happen at the same time.

So, an incremental backup for the changed data only takes a much less time. So what most organizations would tend to do is do a full backup once a week and do incremental backup rest of the day. So, what it means, if you see here, say, on Friday suppose, they do a full backup, then

on Saturday they do an incremental backup, that is, as much as changed from the Friday's full backup.

Sunday they do another incremental backup, which is as much as changed from the Saturday's incremental backup. Monday, they do incremental over Sunday, Tuesday, they do incremental over Monday, and so on. It goes on till Thursday. And again on the next Friday they do a full backup. So, this ensures that most of the time you need a minimum backup window, and, which you can place at a non-peak time and keep the best possible throughput from your system.

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The screenshot shows a presentation slide from IIT Madras. The title is 'Types of Backup Strategies: Incremental Backup (2)'. The slide is part of Module 51, presented by Partha Pratim Das. The left sidebar lists the following topics: Week Recap, Objectives & Outline, What is Backup and Recovery?, Why Backup?, Backup Data Types, Backup Strategies, Full Backup, Incremental Backup, Differential Backup, and Example. The main content area lists the following points:

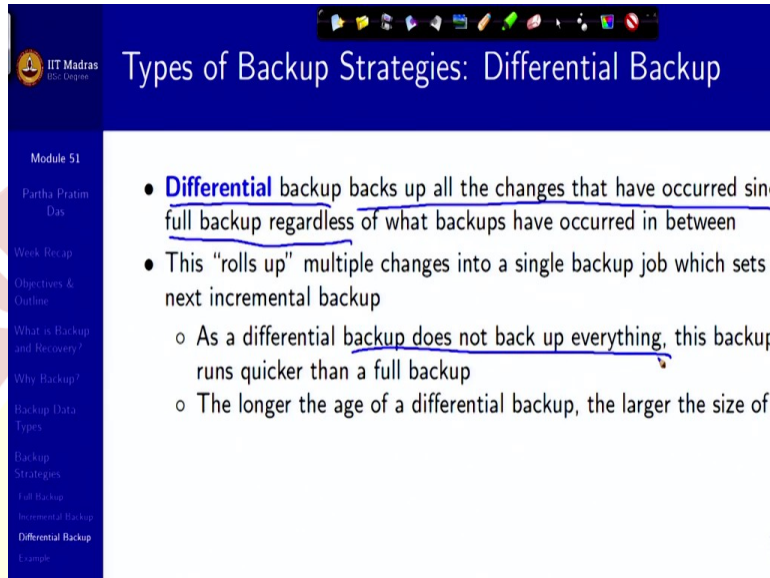
- **Incremental Backup: Advantages**
 - Less storage is used per backup
 - The downtime due to backup is minimized
 - It provides considerable cost reductions over full backups
- **Incremental Backup: Disadvantages**
 - It requires more effort and time during recovery
 - A complete system recovery needs a full backup to start with
 - It cannot be done without the full backups and all incremental ba
 - If any of the intermediate incremental backups are lost, then the r 100%

So, this is a basic idea of the incremental backup. The advantages are obvious. You need less storage, you need less downtime, and it provides a considerable cost reduction over the full backup. Naturally, there are flip side. The flip side is it requires more effort and time during recovery. Because now, with a full backup, one backup will give the entire, I mean, if I had made, every week made a whole copy of my hard disk, then I can, when I need, I can just copy it back. In one go it will be done.

But if I have done increment and only, then every incremental change will have to be brought on with the full backup to start with. So it cannot be done without a full backup, and all incremental backups that have happened from that full backup. So if the system, say, unfortunately goes down on Monday, I have to take the full back of Friday, restore, then the incremental on Saturday, then the incremental on Sunday, and then I will be able to come to the Monday.

So, if any of these intermediate incremental backups are lost, then I cannot restore the system. So, these are, these are some of the disadvantages.

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IIT Madras
Types of Backup Strategies: Differential Backup

Module 51
Partha Pratim Das
Week Recap
Objectives & Outline
What is Backup and Recovery?
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- **Differential** backup backs up all the changes that have occurred since full backup regardless of what backups have occurred in between
- This "rolls up" multiple changes into a single backup job which sets the next incremental backup
 - As a differential backup does not back up everything, this backup runs quicker than a full backup
 - The longer the age of a differential backup, the larger the size of

So we can, you can see that full backup has a very, I mean it is, it is, it is very, dealing the whole system together has a very low granularity. Whereas, incremental backup has a lot more granularity. It is being done in small units every day. So, it has a trade-off between the two. So, differential backup is a strategy that does a medium range granularity.

So, what you do is, differential backup backs up all the changes that occurred with the most recent full backup, irrespective of what other incremental backups that you have taken. You take the last full backup and with respect to that you are doing the differential. So, you can say that doing on a daily basis, incremental is a form of differential.

But it is, differential only has a time window which is more than the incremental backup and less than the full backup. So, it is somewhere in between. So, it, again, does not have to back up everything but can still run the backup much faster than the full backup, and can facilitate the restore, recovery process.

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**IIT Madras**
BSc Degree

Types of Backup Strategies: Differential Backup (2)

Module 51
Partha Pratim Das
Week Recap
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Differential Backup
Example

- To evaluate how differential backups might work within an environment, consider schedule shown in the figure below.

✓ Friday	✓ Saturday	✓ Sunday	✓ Monday	✓ Tuesday	Wednesday	Thursday
Full	Incremental	Incremental	Incremental	Differential	Incremental	Incremental

a) The incremental backup on Saturday backs up all files that have changed since Friday. Likewise all changes since Saturday and Sunday is backed up on Sunday incremental backup respectively.

b) On Tuesday, a differential backup is performed. This backs up all files that have changed since the full backup on Friday. A recovery on Wednesday should only require data from differential backups, **skipping the Saturday/Sunday/Monday incremental backups**.

Recovery on any given day only needs the data from the full backup and the most recent differential backup.



**IIT Madras**
BSc Degree

Types of Backup Strategies: Differential Backup (2)

Module 51
Partha Pratim Das
Week Recap
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Example

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Full	Incremental	Incremental	Incremental	Differential	Incremental	Incremental

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Recovery on any given day only needs the data from the full backup and the most recent differential backup.





IIT Madras
Vellore, Chennai

Module 51

Partha Pratim Das

Week Backup

Objectives & Outline

What is Backup and Recovery?

Why Backup?

Backup Data Types

Backup Strategies

Full Backup

Incremental Backup

Differential Backup

Example

Types of Backup Strategies: Differential Backup (2)

- To evaluate how differential backups might work within an environment, consider schedule shown in the figure below.

Friday	Saturday	Sunday	Monday	Tuesday	Wednesday
Full	Incremental	Incremental	Incremental	Differential	Incremental

a) The incremental backup on Saturday backs up all files that have changed since Friday. Likewise all changes since Saturday and Sunday is backed up on Sunday incremental backup respectively.

b) On Tuesday, a differential backup is performed. This backs up all files that have full backup on Friday. A recovery on Wednesday should only require data from differential backups, **skipping the Saturday/Sunday/Monday incremental backup**.

Recovery on any given day only needs the data from the full backup and the most recent



So, this could be another, a possible scheme. Again, modifying that you again do full backup on Friday, do incremental on Saturday, Sunday, Monday. But on Thursday, you do not do incremental, you do differential. So, what you are doing? This is, this is where you have done full, and then, on, you have just incremental on Saturday. So, it is over the full. You have done incremental on Sunday, it is over the Saturday, incremental on Monday, over the Sunday. On Tuesday, you do a differential.

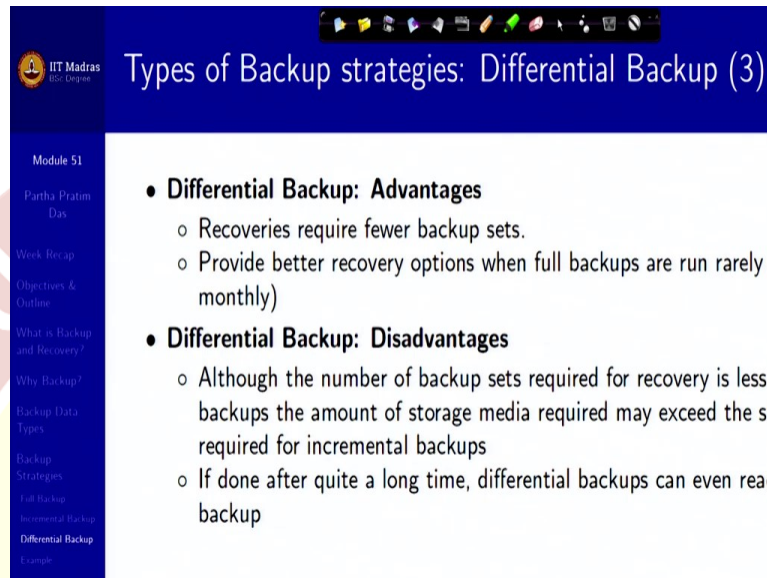
So, what you do, in Tuesday you do not do incremental with respect to Monday. Rather, you do an incremental, differential with respect to Friday. So, which means, and then again when you do the next incremental you do not go back to the last full backup, but rather you go back to the last differential backup. So, when you are doing it on Wednesday, you are not going back to the full backup, rather you are doing it with respect to the differential backup. Then you go on the incremental backup.

So, you have to choose your resolution. Full backup, less granularity, all the advantages, disadvantages, incremental backup, maximum granularity, advantages, disadvantages, and differential backup stands in between to give you a nice trade-off between the two extremes. So, recovery on any given day only needs the data from the full backup and the most recent, recent differential backup and the remaining incremental backup.

So, if you have to restore something on this day, say, Thursday you need the full backup, the differential backup, and just this incremental backup. Whereas, if you did not have the

differential, you would have required, 1, 2, 3, 4, 5, incremental backups. So, that is the basic trade-off. And you choose these time windows based on what your business requirements are.

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100th Degree

Types of Backup strategies: Differential Backup (3)

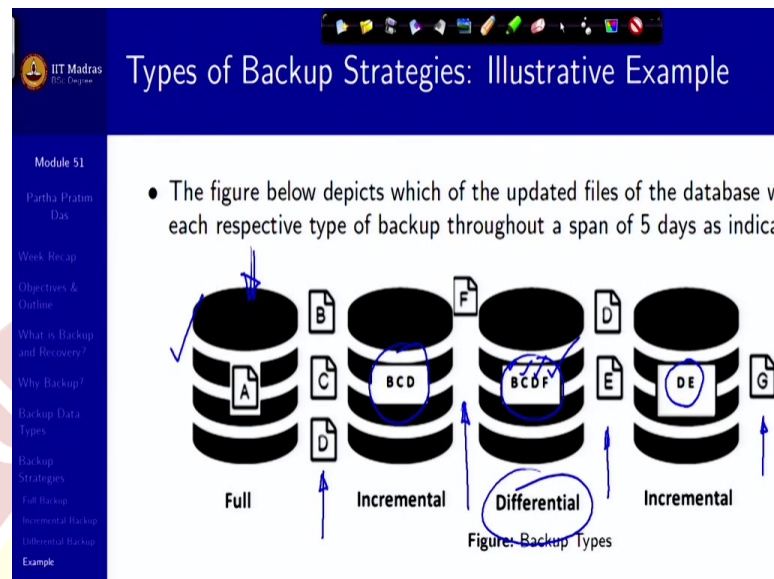
Module 51
Partha Pratim Das
Week Backup
Objectives & Outline
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Example

- **Differential Backup: Advantages**
 - Recoveries require fewer backup sets.
 - Provide better recovery options when full backups are run rarely (monthly)
- **Differential Backup: Disadvantages**
 - Although the number of backup sets required for recovery is less than full backups the amount of storage media required may exceed the storage required for incremental backups
 - If done after quite a long time, differential backups can even reach the size of a full backup

So, advantages of differential backup is the recoveries are required fewer backup sets. So, it is faster, takes less time, gives better recovery option, when the full backup particularly is run less frequently. For example, you might, then with differential, you might run full backup not on a weekly basis but on a monthly basis.

But certainly, it has disadvantages. Although the number of backup sets required for recovery is less, but in differential backup also the amount of storage media required may exceed the storage media for incremental backup because it is a backup over a, after a longer period. It is true that incrementally in between the changes have been recorded. So, you have the recovery options but you are doing it after 3, 4 days so you have a bigger difference, and, so it could even reach the size of a full backup at times. In some systems it does.

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So, this is just an illustration of what you back up on according to this strategy. So, here you are backing up everything, the full backup. After that B, C, D, these three files have changed. So, when you do that incremental, you just backup B, C, D. After that backup, F, file F has changed. So when you do differential, you are doing with respect to the full. So, you do B, C, D as well as the F.

After that D and E has changed, so you do incremental over D, E because you are doing it with respect to this last differential backup. Then only G has changed, but since you are going to do a full backup, you do a backup of everything, not only just G, or no not only does D, E, G. You do the full backup.

So, this is the basic strategy, and you have to find out your time resolution, your cyclicity, what kind of cycles you want, but this is just conceptualized over a span of 5 days. This is how your backups could be strategized.

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Case: Monthly Schedule

Module 51

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Week Recap

Objectives & Outline

What is Backup and Recovery?

Why Backup?

Backup Data Types

Backup Strategies

Full Backup

Incremental Backup

Differential Backup

Example

Case: Monthly Schedule

IIT Madras
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Case: Monthly Data Backup Schedule

Module 51

Partha Pratim Das

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Example

Consider the following backup schedule for a month:

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1/Full	2/Incr	3/Incr	4/Incr	5/Incr	6/Incr	7/Incr
8/Diff	9/Incr	10/Incr	11/Incr	12/Incr	13/Incr	14/Incr
15/Diff	16/Incr	17/Incr	18/Incr	19/Incr	20/Incr	21/Incr
22/Diff	23/Incr	24/Incr	25/Incr	26/Incr	27/Incr	28/Incr
29/Diff	30/Incr	31/Incr				

- Inference
 - Here full backups are performed once per month, but with differentials being p maximum number of backups required for a **complete system recovery** at any backup, **one differential backup, and six incremental backups**
 - A full system recovery will never need more than the full backup from the start differential backup at the start of the relevant week, and the incremental back the week
 - If a policy were used whereby **full backups were done on the first of the month for the rest of the month**, a complete system recovery on last day of month w



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Case: Monthly Data Backup Schedule

Module 51

Partha Pratim Das

Week Backup

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● **Inference**

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So as a, as a case study, we can just look at a very typical monthly backup schedule. For example, let us look at a monthly schedule like this. So, just without, without making things complicated, I have just assumed that the month starts on a Sunday. And Sundays, expecting that the business will be less or the business is closed, more time can be given to backup. So, heavier backups are on Sunday.

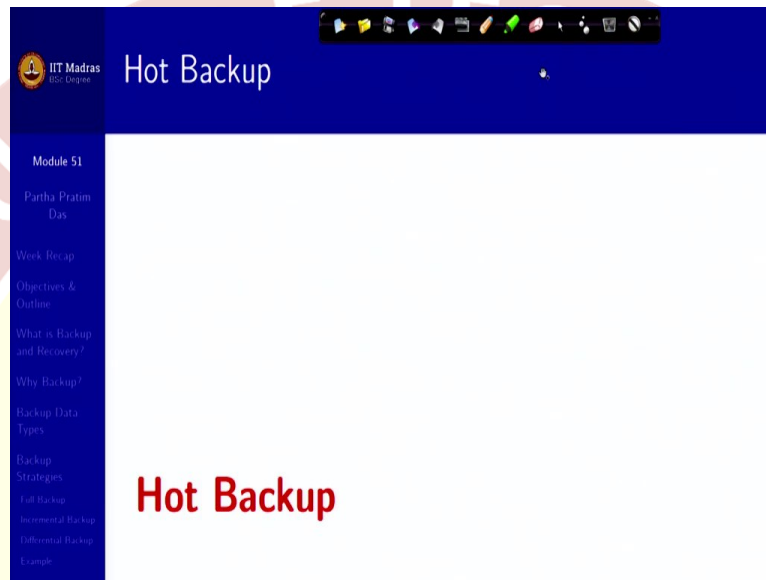
So, what you do on the first of the month, on the Sunday, and if you are choosing Sunday to be preferential, then it may be some date other than the first of the month, also. It may be a Sunday. You do a full backup. And then you keep on doing incremental till Saturday. On the next Sunday, that is, the first Sunday after the full backup, you do a differential. And again do incremental till Saturday.

Again do differential, incremental till Saturday, again differential, incremental till Saturday, again differential. It goes on till you end the month, when you again come and do a full backup. So with this, if you, if with this, this kind of a pattern being followed, what is the maximum number of, maximum number of backups that you will need to restore for, is certainly one full backup, which you took on the first Sunday of the month. Then one differential backup that you took on the last Sunday of the week in which you are trying to recover, trying to restore.

And then you can have at most 6 incremental backups because there are 7 days. So, at most 6 you will require to do, restore the whole system. If you had done full backups on the first Sunday of the month, and incremental for the rest you would have required 31 backup sets to be used.

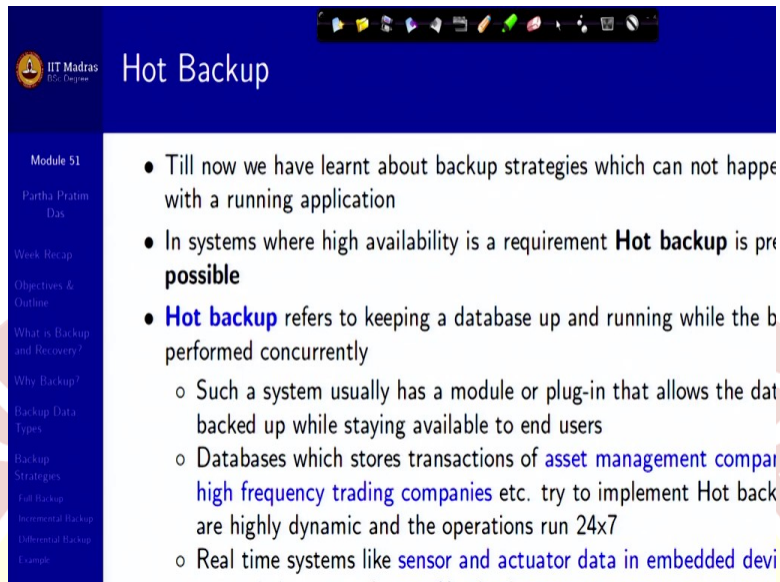
So, there are, there are issues with that. One is there are every incremental backup to be restored has a separate process, separate cost and so on. And further so many files will have to be perfect, so many backup files, again, have to be perfect so that you can actually do the restore. Here, you are left with various different options.

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So, till this point all that we talked of is backing up the data that exist in the disk. And typically this will be called a cold backup. That is, we are saying that we will do the backup when we can take the system off, do the backup put the system on, or maybe just allow the system to run on a minimal skeleton for some urgent work in a limited way, but otherwise you are just backing up.

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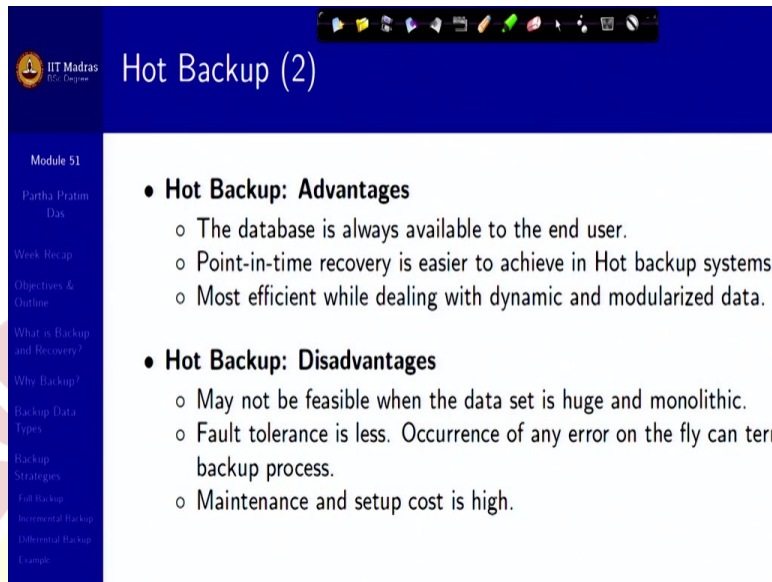
Hot Backup

- Till now we have learnt about backup strategies which can not happen with a running application
- In systems where high availability is a requirement **Hot backup** is possible
- **Hot backup** refers to keeping a database up and running while the backup is performed concurrently
 - Such a system usually has a module or plug-in that allows the database to be backed up while staying available to end users
 - Databases which store transactions of **asset management companies**, **high frequency trading companies** etc. try to implement Hot backup as they are highly dynamic and the operations run 24x7
 - Real time systems like **sensor and actuator data in embedded devices**

But what is important is hot backup, is what if your system goes down when you are, it is actually in operation. Which is, which is quite possible. So, when that happens naturally your database will become inconsistent, and that is the recoverability, or the, that is the cascadeless, or cascaded recovery we were talking of. We were then talking primarily in terms of transactions having to roll back because of some reason.

But if it is to do that for certain reasons which is beyond your control, then you will be in deep difficulty because data might get lost. So, database which stores transactions like asset management companies, hedge funds, stock market transactions need to have hot backups as this data are very dynamic and operations run on 24/7. So, even real time systems will need that.

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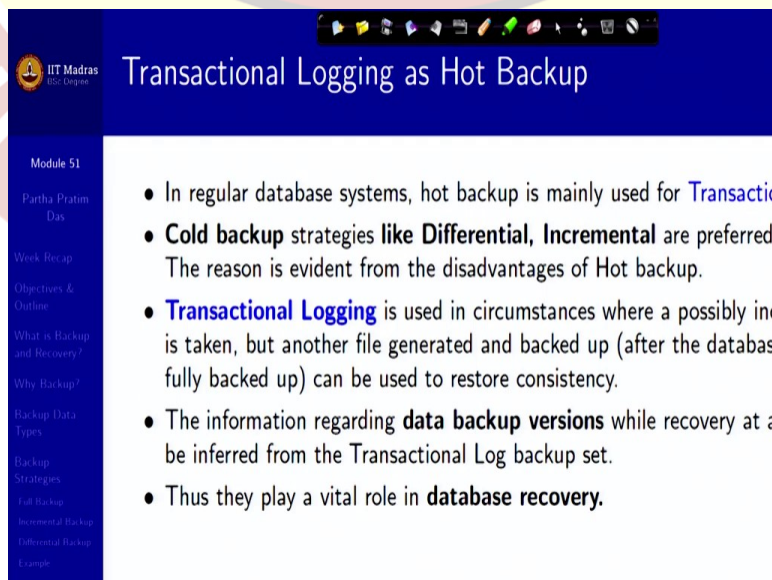
Hot Backup (2)

- **Hot Backup: Advantages**
 - The database is always available to the end user.
 - Point-in-time recovery is easier to achieve in Hot backup systems.
 - Most efficient while dealing with dynamic and modularized data.
- **Hot Backup: Disadvantages**
 - May not be feasible when the data set is huge and monolithic.
 - Fault tolerance is less. Occurrence of any error on the fly can terminate backup process.
 - Maintenance and setup cost is high.

So, hot backups are mechanisms by which the database always remains available to the end user, and the point of time recovery, point in time recovery is easier in this system, and certainly it is most efficient when you deal with dynamic and modularized data.

Disadvantages, of course are there. It is may not be feasible when the data set is huge or monolithic. That is the reason you have to make the database very typically modular. Fault tolerance is less. That is, if something fails then detecting that on the fly can be difficult. So, any error could terminate your whole backup process. Maintenance and setup cost is large and so on.

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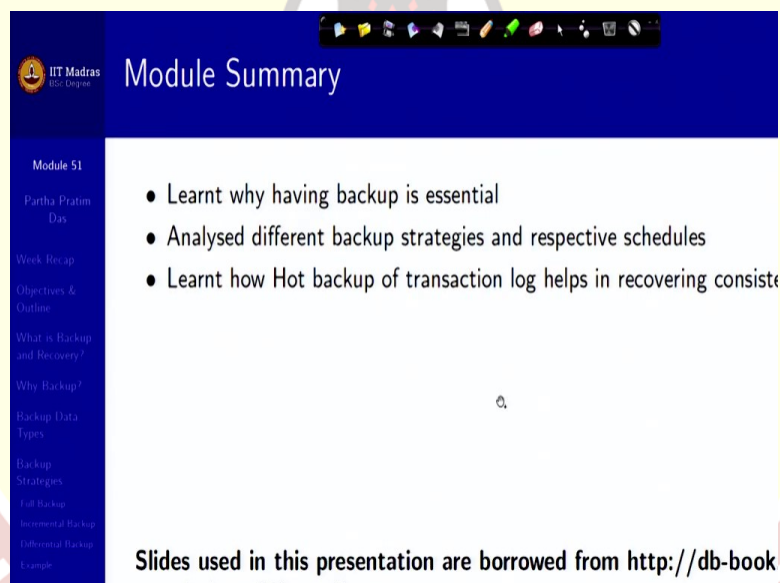
Transactional Logging as Hot Backup

- In regular database systems, hot backup is mainly used for **Transactional**
- **Cold backup** strategies like **Differential**, **Incremental** are preferred. The reason is evident from the disadvantages of Hot backup.
- **Transactional Logging** is used in circumstances where a possibly inc is taken, but another file generated and backed up (after the database fully backed up) can be used to restore consistency.
- The information regarding **data backup versions** while recovery at a be inferred from the Transactional Log backup set.
- Thus they play a vital role in **database recovery**.

So typically, the hot backup use the concept of transaction logs, transaction log backup. Transaction logging is a mechanism, and that is what we are going to discuss in the coming modules, is a mechanism by which while you do the transactions you are doing read-write, commit, abort and all that, you keep a copy of what you have done, you keep a ledger in terms of what you have done.

So, the transaction log usually is much lighter compared to the actual database. And you can have points where the transaction log is consistent with the database completely, and therefore you do not need the previous log anymore. So, how you use this transaction log in hot backup, and what is transaction logging is going to be the next thing that we are going to discuss while we talk about the recovery in transactions.

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Module Summary

Module 51
Partha Pratim Das

- Learnt why having backup is essential
- Analysed different backup strategies and respective schedules
- Learnt how Hot backup of transaction log helps in recovering consistency

Slides used in this presentation are borrowed from <http://db-book>

So, this brings us to the end of this module. Thank you for your attention, and we will continue in the next module.